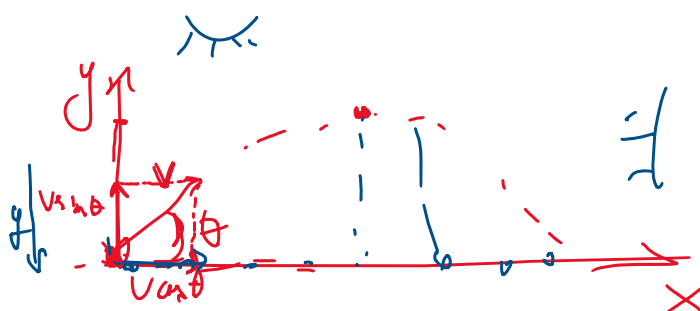
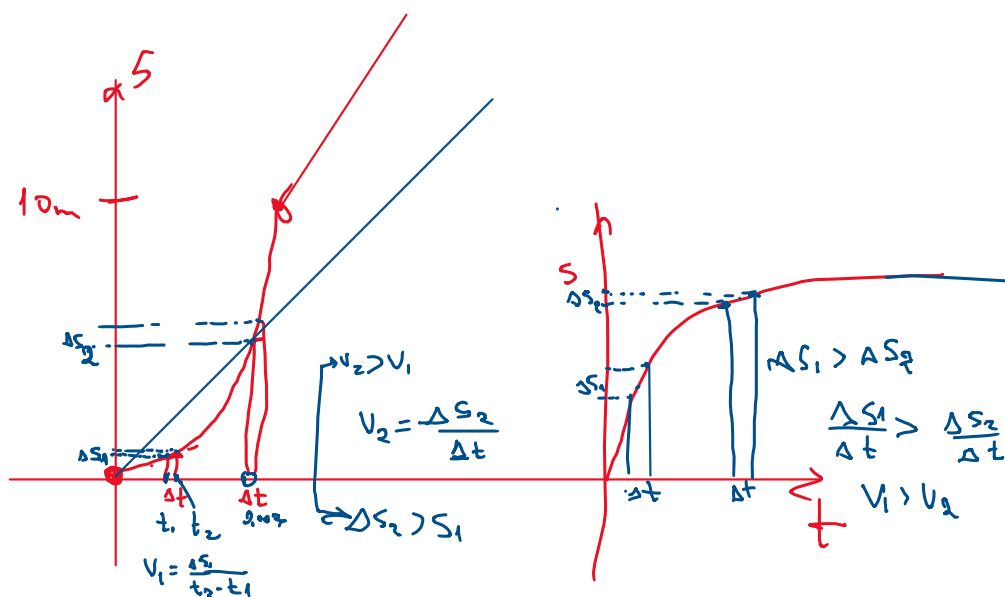
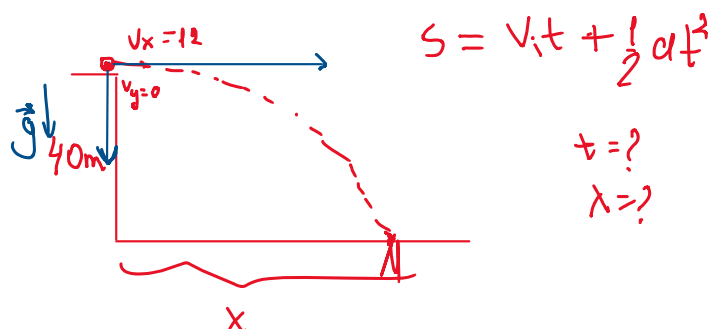


- 6 A girl is riding a bicycle at a constant velocity of 3.0 m s^{-1} along a straight road. At time $t = 0$, she passes a boy sitting on a stationary bicycle. At time $t = 0$, the boy sets off to catch up with the girl. His velocity increases from time $t = 0$ until $t = 5.0 \text{ s}$, when he has covered a distance of 10 m . He then continues at a constant velocity of 4.0 m s^{-1} .
- Draw the displacement-time graph for the girl from $t = 0$ to 12 s .
 - On the same graph axes, draw the displacement-time graph for the boy.
 - Using your graph, determine the value of t when the boy catches up with the girl.



A stone is thrown horizontally with a velocity of 12 m s^{-1} from the top of a vertical cliff.

Calculate how long the stone takes to reach the ground 40 m below and how far the stone lands from the base of the cliff.



$$40 = \frac{gt^2}{2} \quad t^2 = 8 \quad t = \sqrt{8}$$

$$S = 12 \cdot \sqrt{8} +$$

A ball is thrown with an initial velocity of 20 m s^{-1} at an angle of 30° to the horizontal (Figure 2.32). Calculate the horizontal distance travelled by the ball (its range).

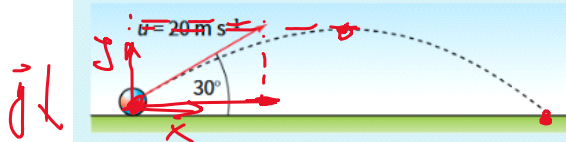


Figure 2.32 Where will the ball land?

$$V_x = V \cos 30^\circ = V \frac{\sqrt{3}}{2} = 17,3 \text{ m s}^{-1}$$

$$V_y = \frac{V}{2} = 10 \text{ m s}^{-1}$$

$$V_y = V_i + a t$$

$$0 = 10 - g t$$

$$t_m = \frac{10}{9,8} = 1,02$$

$$t = 2,04$$

$$s = V_x \cdot t$$

$$s_y = 0 = V_{iy} t - \frac{gt^2}{2}$$

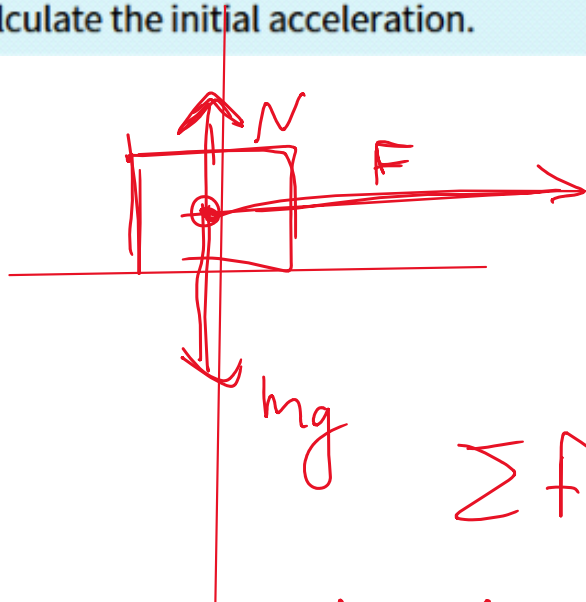
$$0 = 10t - 5t^2$$

$$t = 0$$

$$t = 2,04$$

$$F = m a$$

A cyclist of mass 60 kg rides a bicycle of mass 20 kg . When starting off, the cyclist provides a force of 200 N . Calculate the initial acceleration.



$$\sum F_i = 0$$

$$\downarrow y \quad mg - N = 0 \quad mg = N$$

An object will remain at rest or in a state of uniform motion unless it is acted on by a resultant force.



A car of mass 500 kg is travelling along a flat road. The forward force provided between the car tyres and the road is 300 N and the air resistance is 200 N. Calculate the acceleration of the car.

The maximum forward force a car can provide is 500 N. The air resistance F which the car experiences depends on its speed according to $F = 0.2v^2$, where v is the speed in m s^{-1} . Determine the top speed of the car.



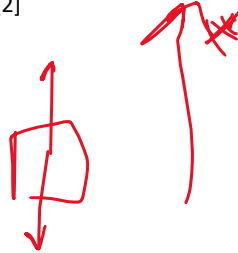


When two bodies interact, the forces they exert on each other are equal in magnitude and opposite in direction.

The mass of a spacecraft is 70 kg. As the spacecraft takes off from the Moon, the upwards force on the spacecraft caused by the engines is 500 N. The gravitational field strength on the Moon is 1.6 N kg^{-1} .

Determine:

- the weight of the spacecraft on the Moon [2]
- the resultant force on the spacecraft [2]
- the acceleration of the spacecraft.



$3 \mu\text{m}$ 5 ms

Determine the speed in m s^{-1} of an object that travels:

- $3 \mu\text{m}$ in 5 ms [2]
- 6 km in 3 Ms [2]
- 8 pm in 4 ns .

A metal ball is dropped into a tall cylinder of oil. The ball initially accelerates but soon reaches a terminal velocity.

- By considering the forces on the metal ball bearing, explain why it first accelerates but then reaches terminal velocity. [3]
- Describe how you would show that the metal ball reaches terminal velocity.

Figure 3.19 shows the speed–time graphs for two falling balls.

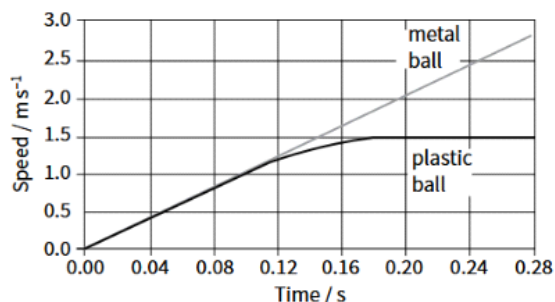
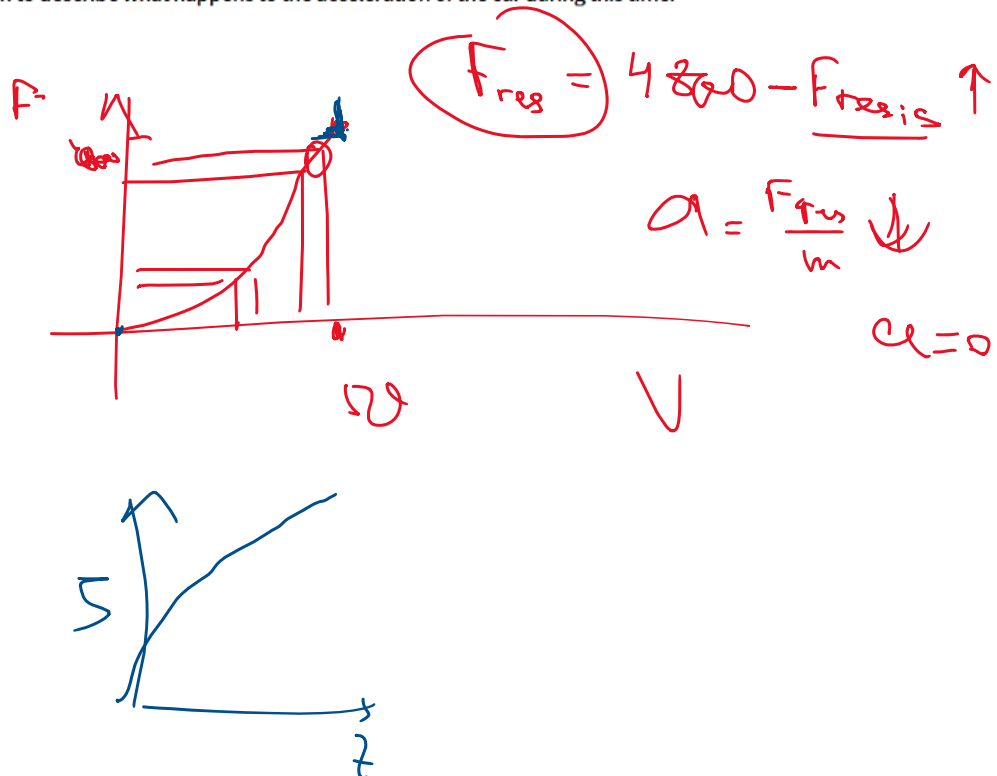


Figure 3.19 For End-of-chapter Question 7.

- Determine the terminal velocity of the plastic ball.
- Both balls are of the same size and shape but the metal ball has a greater mass. Explain, in terms of Newton's laws of motion and the forces involved, why the plastic ball reaches a constant velocity but the metal ball does not.
- Explain why both balls have the same initial acceleration.

- 8 A car of mass 1200 kg accelerates from rest to a speed of 8.0 m s^{-1} in a time of 2.0 s.
- Calculate the forward driving force acting on the car while it is accelerating. Assume that, at low speeds, all frictional forces are negligible.
 - At high speeds the resistive frictional force F produced by air on a body moving with velocity v is given by the equation $F = bv^2$, where b is a constant.
 - Derive the base units of force in the SI system.
 - Determine the base units of b in the SI system.
 - The car continues with the same forward driving force and accelerates until it reaches a top speed of 50 m s^{-1} . At this speed the resistive force is given by the equation $F = bv^2$. Determine the value of b for the car.
 - Sketch a graph showing how the value of F varies with v over the range 0 to 50 m s^{-1} . Use your graph to describe what happens to the acceleration of the car during this time.



- 5 Figure 3.18 shows a man who is just supporting the weight of a box. Two of the forces acting are shown in the diagram. According to Newton's third law, each of these forces is paired with another force.

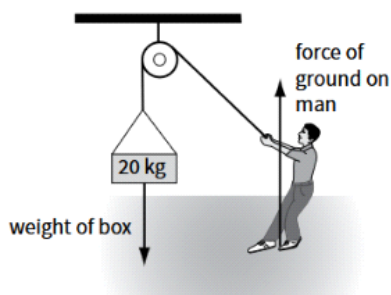
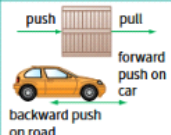
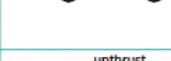
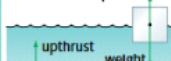


Figure 3.18 For End-of-chapter Question 5.

For a the weight of the box and b the force of the ground on the man, state:

- the body that the other force acts upon
- the direction of the other force
- the type of force involved.

Diagram	Force	Important situations
	Pushes and pulls. You can make an object accelerate by pushing and pulling it. Your force is shown by an arrow pushing (or pulling) the object. The engine of a car provides a force to push backwards on the road.	■ pushing and pulling ■ lifting ■ force of car engine

Diagram	Force	Important situations
	Pushes and pulls. You can make an object accelerate by pushing and pulling it. Your force is shown by an arrow pushing (or pulling) the object. The engine of a car provides a force to push backwards on the road. Frictional forces from the road on the tyre push the car forwards.	- pushing and pulling - lifting - force of car engine - attraction and repulsion by magnets and by electric charges
	Weight. This is the force of gravity acting on the object. It is usually shown by an arrow pointing vertically downwards from the object's centre of gravity.	- any object in a gravitational field - less on the Moon
	Friction. This is the force which arises when two surfaces rub over one another. If an object is sliding along the ground, friction acts in the opposite direction to its motion. If an object is stationary, but tending to slide – perhaps because it is on a slope – the force of friction acts up the slope to stop it from sliding down. Friction always acts along a surface, never at an angle to it.	- pulling an object along the ground - vehicles cornering or skidding - sliding down a slope
	Drag. This force is similar to friction. When an object moves through air, there is friction between it and the air. Also, the object has to push aside the air as it moves along. Together, these effects make up drag. Similarly, when an object moves through a liquid, it experiences a drag force. Drag acts to oppose the motion of an object; it acts in the opposite direction to the object's velocity. It can be reduced by giving the object a streamlined shape.	- vehicles moving - aircraft flying - parachuting - objects falling through air or water - ships sailing
	Upthrust. Any object placed in a fluid such as water or air experiences an upwards force. This is what makes it possible for something to float in water. Upthrust arises from the pressure which a fluid exerts on an object. The deeper you go, the greater the pressure. So there is more pressure on the lower surface of an object than on the upper surface, and this tends to push it upwards. If upthrust is greater than the object's weight, it will float up to the surface.	- boats and icebergs floating - people swimming - divers surfacing - a hot air balloon rising
	Contact force. When you stand on the floor or sit on a chair, there is usually a force which pushes up against your weight, and which supports you so that you do not fall down. The contact force is sometimes known as the normal reaction of the floor or chair. (In this context, normal means 'perpendicular'). The contact force always acts at right angles to the surface which produces it. The floor pushes straight upwards; if you lean against a wall, it pushes back against you horizontally.	- standing on the ground - one object sitting on top of another - leaning against a wall - one object bouncing off another
	Tension. This is the force in a rope or string when it is stretched. If you pull on the ends of a string, it tends to stretch. The tension in the string pulls back against you. It tries to shorten the string. Tension can also act in springs. If you stretch a spring, the tension pulls back to try to shorten the spring. If you squash (compress) the spring, the tension acts to expand the spring.	- pulling with a rope - squashing or stretching a spring

- A motorway designer can assume that cars approaching a motorway enter a slip road with a velocity of 10 m s^{-1} and reach a velocity of 30 m s^{-1} before joining the motorway. Calculate the minimum length for the slip road, assuming that vehicles have an acceleration of 4.0 m s^{-2} .
- A train is travelling at 50 m s^{-1} when the driver applies the brakes and gives the train a constant deceleration of magnitude 0.50 m s^{-2} for 100 s. Describe what happens to the train. Calculate the distance travelled by the train in 100 s.
- A boy stands on a cliff edge and throws a stone vertically upwards at time $t = 0$. The stone leaves his hand at 20 m s^{-1} . Take the acceleration of the ball as 9.81 m s^{-2} .
 - Show that the equation for the displacement of the ball is:

$$s = 20t - 4.9t^2$$
 - What is the height of the stone 2.0 s after release and 6.0 s after release?
 - When does the stone return to the level of the boy's hand? Assume the boy's hand does not move vertically after the ball is released.

- 4 The graph in Figure 2.33 shows the variation of velocity with time of two cars, A and B, which are travelling in the same direction over a period of time of 40 s. Car A, travelling at a constant velocity of 40 m s^{-1} , overtakes car B at time $t = 0$. In order to catch up with car A, car B immediately accelerates uniformly for 20 s to reach a constant velocity of 50 m s^{-1} . Calculate:

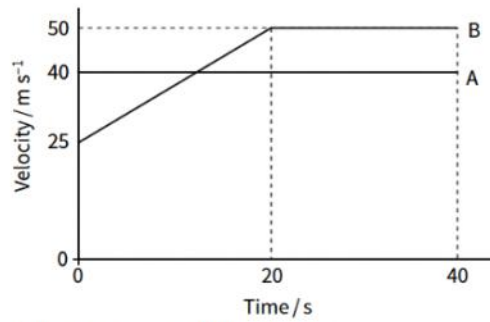


Figure 2.33 Velocity-time graphs for two cars, A and B. For End-of-chapter Question 4.

- how far A travels during the first 20 s
 - the acceleration and distance of travel of B during the first 20 s
 - the additional time taken for B to catch up with A
 - the distance each car will have then travelled since $t = 0$.
- 5 An athlete competing in the long jump leaves the ground with a velocity of 5.6 m s^{-1} at an angle of 30° to the horizontal.
- Determine the vertical component of the velocity and use this value to find the time between leaving the ground and landing.
 - Determine the horizontal component of the velocity and use this value to find the horizontal distance travelled.
- 6 Figure 2.34 shows an arrangement used to measure the acceleration of a metal plate as it falls vertically. The metal plate is released from rest and falls a distance of 0.200 m before breaking light beam 1. It then falls a further 0.250 m before breaking light beam 2.

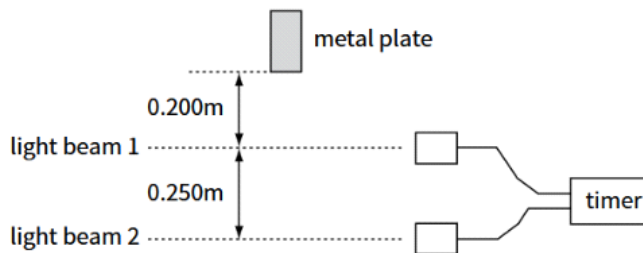


Figure 2.34 For End-of-chapter Question 6.

- Calculate the time taken for the plate to fall 0.200 m from rest. (You may assume that the metal plate falls with an acceleration equal to the acceleration of free fall.)
- The timer measures the speed of the metal plate as it falls through each light beam. The speed as it falls through light beam 1 is 1.92 m s^{-1} and the speed as it falls through light beam 2 is 2.91 m s^{-1} .
 - Calculate the acceleration of the plate between the two light beams.
 - State and explain one reason why the acceleration of the plate is not equal to the acceleration of free fall.

- 7 Figure 2.35 shows the velocity–time graph for a vertically bouncing ball. The ball is released at A and strikes the ground at B. The ball leaves the ground at D and reaches its maximum height at E. The effects of air resistance can be neglected.

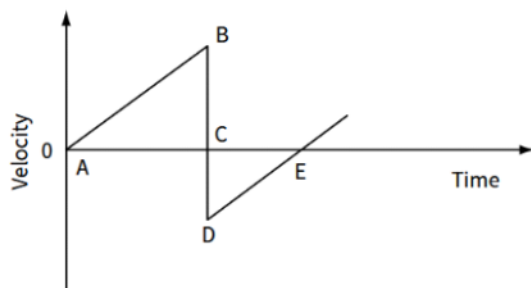


Figure 2.35 For End-of-chapter Question 7.

- a State:
- why the velocity at D is negative
 - why the gradient of the line AB is the same as the gradient of line DE
 - what is represented by the area between the line AB and the time axis
 - why the area of triangle ABC is greater than the area of triangle CDE.
- b The ball is dropped from rest from an initial height of 1.2 m. After hitting the ground the ball rebounds to a height of 0.80 m. The ball is in contact with the ground between B and D for a time of 0.16 s. Using the acceleration of free fall, calculate:
- the speed of the ball immediately before hitting the ground
 - the speed of the ball immediately after hitting the ground
 - the acceleration of the ball while it is in contact with the ground. State the direction of this acceleration.
- 8 A student measures the speed v of a trolley as it moves down a slope. The variation of v with time t is shown in the graph in Figure 2.36.

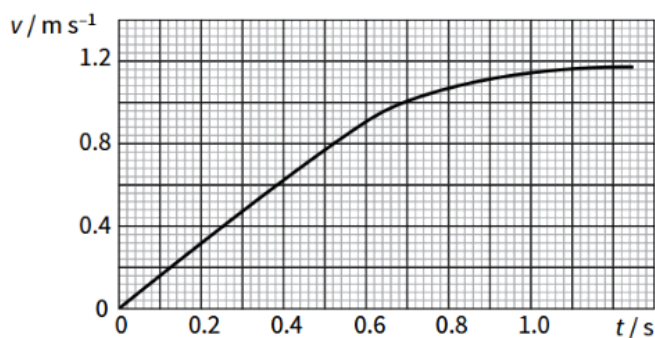


Figure 2.36 For End-of-chapter Question 8.

- Use the graph to find the acceleration of the trolley when $t = 0.7$ s.
- State how the acceleration of the trolley varies between $t = 0$ and $t = 1.0$ s. Explain your answer by reference to the graph.
- Determine the distance travelled by the trolley between $t = 0.6$ and $t = 0.8$ s.
- The student obtained the readings for v using a motion sensor. The readings may have random errors and systematic errors. Explain how these two types of error affect the velocity–time graph.

- 9 A car driver is travelling at speed v on a straight road. He comes over the top of a hill to find a fallen tree on the road ahead. He immediately brakes hard but travels a distance of 60 m at speed v before the brakes are applied. The skid marks left on the road by the wheels of the car are of length 140 m (Figure 2.37). The police investigate whether the driver was speeding and establish that the car decelerates at 2.0 m s^{-2} during the skid.

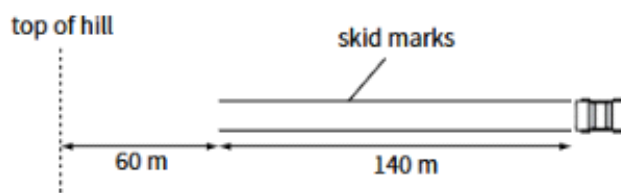


Figure 2.37 For End-of-chapter Question 9.

- Determine the initial speed v of the car before the brakes are applied.
 - Determine the time taken between the driver coming over the top of the hill and applying the brakes. Suggest whether this shows whether the driver was alert to the danger.
 - The speed limit on the road is 100 km/h. Determine whether the driver was breaking the speed limit.
- 10 A hot-air balloon rises vertically. At time $t = 0$, a ball is released from the balloon. Figure 2.38 shows the variation of the ball's velocity v with t . The ball hits the ground at $t = 4.1 \text{ s}$.

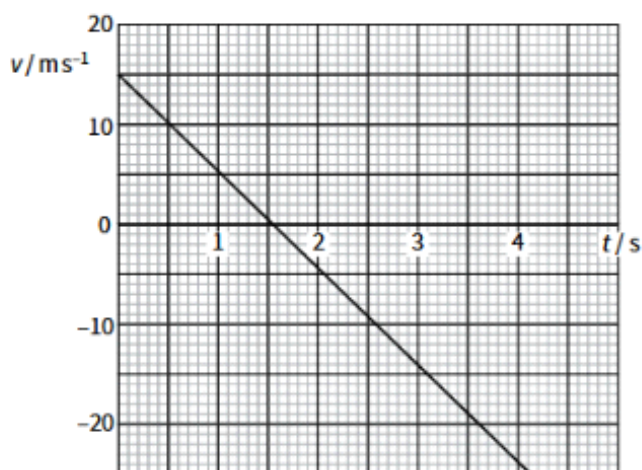


Figure 2.38 For End-of-chapter Question 10.

- Explain how the graph shows that the acceleration of the ball is constant.
 - Use the graph to:
 - determine the time at which the ball reaches its highest point
 - show that the ball rises for a further 12 m between release and its highest point
 - determine the distance between the highest point reached by the ball and the ground.
 - The equation relating v and t is $v = 15 - 9.81t$. Explain the significance in the equation of:
 - the number 15
 - the negative sign.
- 11 An aeroplane is travelling horizontally at a speed of 80 m s^{-1} and drops a crate of emergency supplies (Figure 2.39). To avoid damage, the maximum vertical speed of the crate on landing is 20 m s^{-1} . You may assume air resistance is negligible.



Figure 2.39 For End-of-chapter Question 11.

- Calculate the maximum height of the aeroplane when the crate is dropped.

- 11** An aeroplane is travelling horizontally at a speed of 80 m s^{-1} and drops a crate of emergency supplies (Figure 2.39). To avoid damage, the maximum vertical speed of the crate on landing is 20 m s^{-1} . You may assume air resistance is negligible.



Figure 2.39 For End-of-chapter Question 11.

- a** Calculate the maximum height of the aeroplane when the crate is dropped.
- b** Calculate the time taken for the crate to reach the ground from this height.
- c** The aeroplane is travelling at the maximum permitted height. Calculate the horizontal distance travelled by the crate after it is released from the aeroplane.